Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
Fall 2026

Recitation 3 Solutions

Problem 3.1

Of the students in a class, 10 are freshmen, 30 are sophomores, 20 are juniors and 10 are seniors. Let X be the year of a student chosen at random in the class.

(a) Determine the alphabet $\mathcal{X}$ and the PMF p_X for the random variable X.

Solution:

Based on the problem setup, X takes on values $1, 2, 3, 4$ and therefore $\mathcal{X} = \{1, 2, 3, 4\}$. Since the probability of a randomly chosen student to be from a certain year is proportional to the number of students from that year, we have

$$p_X(x) = \begin{cases} 10\,\alpha & x = 1, 4 \\ 30\,\alpha & x = 2 \\ 20\,\alpha & x = 3, \end{cases}$$

where α is a constant that ensures that the PMF of X sums to one:

$$\begin{aligned} 10\,\alpha + 30\,\alpha + 20\,\alpha + 10\,\alpha &= 1 \\ \alpha &= 1/70, \end{aligned}$$

and

$$p_X(x) = \begin{cases} 1/7 & x = 1, 4 \\ 3/7 & x = 2 \\ 2/7 & x = 3. \end{cases}$$

(b) Determine the mean, variance, and entropy of X.

Solution:

We have

$$\begin{aligned} \mathbb{E}[X] &= \frac{1}{7} \cdot 1 + \frac{3}{7} \cdot 2 + \frac{2}{7} \cdot 3 + \frac{1}{7} \cdot 4 = \frac{17}{7} \approx 2.43 \\ \mathbb{E}[X^2] &= \frac{1}{7} \cdot 1 + \frac{3}{7} \cdot 4 + \frac{2}{7} \cdot 9 + \frac{1}{7} \cdot 16 = \frac{47}{7} \approx 6.71 \\ \sigma_X^2 &\triangleq \operatorname{var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{47}{7} - \left(\frac{17}{7}\right)^2 = \frac{40}{49} \approx 0.82 \\ \sigma_X &= \sqrt{\operatorname{var}(X)} \approx 0.9 \\ H(X) &= -\left[2 \cdot \frac{1}{7} \cdot \log_2 \frac{1}{7} + \frac{3}{7} \cdot \log_2 \frac{3}{7} + \frac{2}{7} \cdot \log_2 \frac{2}{7}\right] \approx 1.84 \text{ bits.} \end{aligned}$$

Let's compare this to a uniformly distributed random variable Y over the same alphabet $\mathcal{X} = \{1, 2, 3, 4\}$, for which

$$\begin{aligned}
p_Y(y) &= 1/4, \qquad \text{for } y = 1, 2, 3, 4, \\
\mathbb{E}[Y] &= \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 3 + \frac{1}{4} \cdot 4 = 2.5 \\
\mathbb{E}[Y^2] &= \frac{1}{4} \cdot 1 + \frac{1}{4} \cdot 4 + \frac{1}{4} \cdot 9 + \frac{1}{4} \cdot 16 = 7.5 \\
\sigma_Y^2 &\triangleq \operatorname{var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = 1.25 \\
\sigma_Y &= \sqrt{\operatorname{var}(Y)} \approx 1.1 \\
H(Y) &= \log_2 |\mathcal{X}| = 2 \,\text{bits}.
\end{aligned}$$

Since the distribution of X is skewed towards the lower values, its mean is lower than that of a uniform distribution. Not surprisingly, the entropy of X is lower than that of the uniform random variable Y. The variance of X is also lower than the variance of Y, but it is not universally true that a uniform distribution achieves the highest variance.

(c) Suppose each freshman takes 4 courses each term, and all other students take 5 courses each term. Determine the expected number of courses a student chosen at random from the class is taking this semester.

Solution:

We define random variable $Y \in \{4, 5\}$ to be the number of courses a student chosen at random takes. We can express Y as a function of X as follows:

$$Y = 4 + \mathbb{1}(X > 1),$$

where $\mathbb{1}(\cdot)$ is the indicator function.

We can either derive the distribution

$$p_Y(y) = \begin{cases} 1/7 & y = 4 \\ 6/7 & y = 5, \end{cases}$$

and compute

$$\mathbb{E}[Y] = \frac{1}{7} \cdot 4 + \frac{6}{7} \cdot 5 = \frac{34}{7},$$

or use the linearity of expectation to directly obtain

$$\begin{aligned}
\mathbb{E}[Y] &= \mathbb{E}[4 + \mathbb{1}(X > 1)] \\
&= \mathbb{E}[4] + \mathbb{E}[\mathbb{1}(X > 1)] \\
&= 4 + \mathbb{P}(X > 1) \\
&= 4 + 1 - p_X(1) \\
&= \frac{34}{7}.
\end{aligned}$$

(d) If 10 more seniors join the class, will the entropy of X increase of decrease?

Solution:

The entropy will increase. Intuitively, the new PMF is closer to the uniform distribution and therefore its entropy is higher than that of the original distribution. Formally, let Z be the random variable that represents the year of a student chosen at random in this new set of 80 students. We have

$$p_Z(z) = \begin{cases} 1/8 & z = 1 \\ 3/8 & z = 2 \\ 1/4 & z = 3, 4 \end{cases}$$

and

$$H(Z) = -\left[\frac{1}{8} \cdot \log_2 \frac{1}{8} + \frac{3}{8} \cdot \log_2 \frac{3}{8} + 2 \cdot \frac{1}{4} \cdot \log_2 \frac{1}{4}\right] \approx 1.91 \,\text{bits} > H(X).$$

Remark: Interestingly, if more seniors continue to join the class, at some point the distribution will get further away from the uniform distribution and its entropy will start decreasing. As an example, if 40 more seniors join the class (bringing the total number of students to 120), the entropy of the resulting distribution can be expressed as

$$H(Z') = -\left[\frac{1}{12} \cdot \log_2 \frac{1}{12} + \frac{1}{4} \cdot \log_2 \frac{1}{4} + + \frac{1}{6} \cdot \log_2 \frac{1}{6} + \frac{1}{2} \cdot \log_2 \frac{1}{2}\right] \approx 1.73 \,\text{bits}.$$

As a fun exercise, you could write code to compute and plot the entropy of the distribution as a function of the number of seniors in the class.

Problem 3.2

Bob has a biased coin that turns up heads with probability p. He tosses the coin repeatedly until he observes heads. Assume that coin tosses are independent of each other. Let X be the number of coin tosses up to and including the toss that produced heads.

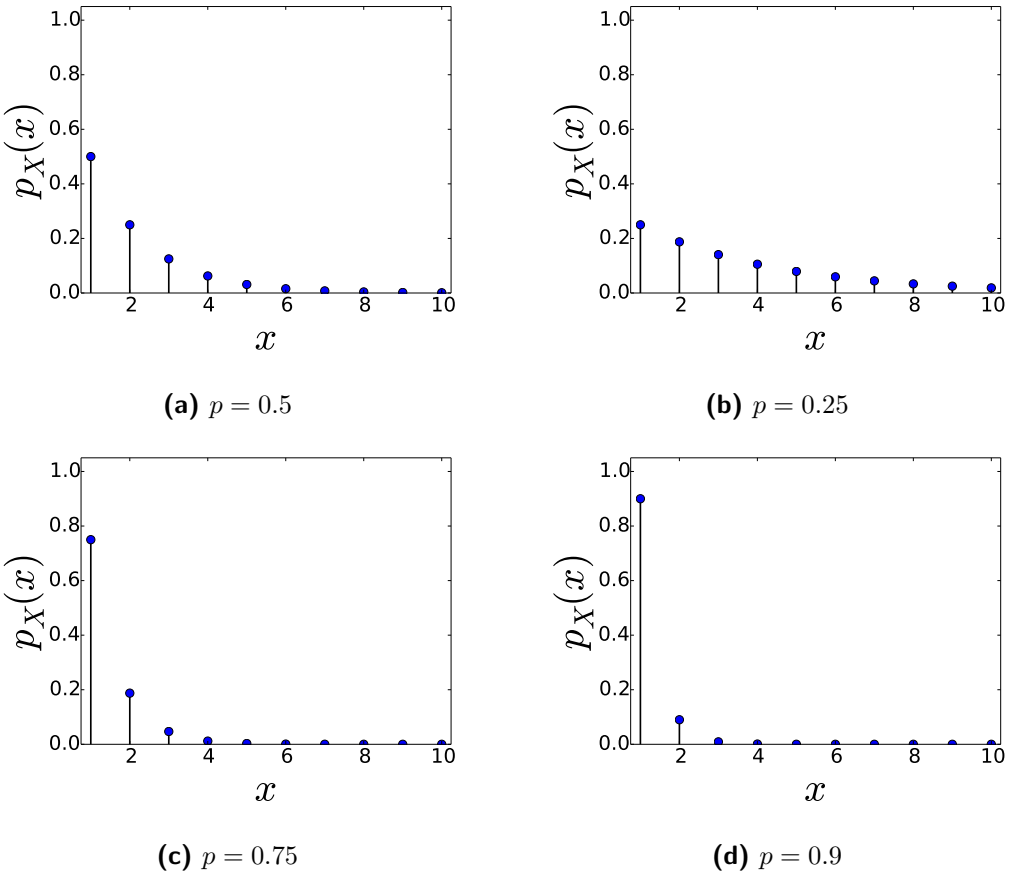
(a) Determine and sketch $p_X(x)$ for $x = 1, 2, \ldots$.

Solution:

Because the coin tosses are independent, we obtain

$$\begin{aligned} p_X(x) &= \mathbb{P}(\texttt{T} \text{ occurs } x-1 \text{ times, followed by } \texttt{H}) \\ &= \underbrace{\mathbb{P}(\texttt{T}) \cdot \cdots \cdot \mathbb{P}(\texttt{T})}_{x-1 \text{ times}} \cdot \mathbb{P}(\texttt{H}) \qquad \text{(independence of coin tosses)} \\ &= \underbrace{(1-p) \cdot \cdots \cdot (1-p)}_{x-1 \text{ times}} \cdot p \\ &= (1-p)^{x-1} p, \quad \text{for } x = 1, 2, \ldots \end{aligned}$$

Recall that this is a geometric distribution, so called because of the geometric nature of the decay.



(a) $p = 0.5$ **(b)** $p = 0.25$

(c) $p = 0.75$ **(d)** $p = 0.9$

Figure 1: Geometric PMF for different values of p

What does the geometric distribution look like? We can write a little bit of code to plot this for a number of values of p. The following code achieves this:

```
# Plot the geometric distribution for various values of of p
import matplotlib
matplotlib.use('Agg')

from scipy.stats import geom
import matplotlib.pyplot as plt
import numpy as np

p_vals = [0.5,0.25,0.75,0.9]
x = np.arange(10) + 1

plt.figure(1,dpi=300,figsize=[8,6])

for k in np.arange(len(p_vals)):
```

```
p = p_vals[k]
plt.plot(x,geom.pmf(x,p),'bo',markersize=10)
plt.vlines(x,[0],geom.pmf(x,p),linewidth=2)
plt.axis([0.75,10.24,0,1.05])
plt.xlabel('$x$',fontsize=48)
plt.ylabel('$p_{X}(x)$',fontsize=48)
plt.tick_params(axis='both',which='major',labelsize=24)
fname = 'geometricpdf_%d.pdf'%k
plt.savefig(fname,bbox_inches='tight')
plt.cla()
```

The plots produced by this code are shown in Fig. 1 above.

We see that $p_X(1) = p$, and then the probability falls off geometrically. We only plot up to $x = 10$ in this example.

(b) Show that $\sum_{x=1}^{\infty} p_X(x) = 1$.

Hint: Recall that for $|r| < 1$,

$$\sum_{i=0}^{\infty} r^i = \frac{1}{1-r},$$

which is referred to as a geometric series.

Solution:

Using the geometric series formula, we have

$$\sum_{x=1}^{\infty}(1-p)^{x-1}p = p\sum_{i=0}^{\infty}(1-p)^i = p\,\frac{1}{1-(1-p)} = p\frac{1}{p} = 1.$$

(c) Show that for any random variable $Y \in \{0, 1, 2, \ldots\}$, $\mathbb{E}[Y] = \sum_{y=1}^{\infty} \mathbb{P}(Y \geq y)$.

Solution:

$$\begin{aligned}
\mathbb{E}[Y] &= \sum_{y=0}^{\infty} y\, p_Y(y) \\
&= 0 \cdot p_Y(0) + 1 \cdot p_Y(1) + 2 \cdot p_Y(2) + 3 \cdot p_Y(3) + 4 \cdot p_Y(4) + \cdots \\
&= p_Y(1) + \big[p_Y(2) + p_Y(2)\big] + \big[p_Y(3) + p_Y(3) + p_Y(3)\big] \\
&\quad + \big[p_Y(4) + p_Y(4) + p_Y(4) + p_Y(4)\big] + \cdots
\end{aligned}$$

$$
\begin{aligned}
&= \underbrace{\big[p_Y(1) + p_Y(2) + p_Y(3) + p_Y(4) + \cdots\big]}_{1-p_Y(0)=\mathbb{P}(Y\geq 1)} \\
&\quad + \underbrace{\big[p_Y(2) + p_Y(3) + p_Y(4) + \cdots\big]}_{1-p_Y(0)-p_Y(1)=\mathbb{P}(Y\geq 2)} + \big[p_Y(3) + p_Y(4) + \cdots\big] + \cdots \\
&= \mathbb{P}(Y \geq 1) + \mathbb{P}(Y \geq 2) + \mathbb{P}(Y \geq 3) + \cdots \\
&= \sum_{y=1}^{\infty} \mathbb{P}(Y \geq y).
\end{aligned}
$$

Using a more compact notation (good practice for working with summation notation), we have

$$
\mathbb{E}[Y] = \sum_{y=0}^{\infty} y\, p_Y(y) = \sum_{y=1}^{\infty} \sum_{k=1}^{y} p_Y(y) = \sum_{k=1}^{\infty} \sum_{y=k}^{\infty} p_Y(y) = \sum_{k=1}^{\infty} \mathbb{P}(Y \geq k).
$$

(d) Show that $\mathbb{E}[X] = 1/p$.

Hint: You may find the result in part (c) helpful.

Solution:

Event $\{X \geq x\}$ indicates that the first heads happens on at least x-th toss, which is equivalent to saying that the first $x-1$ tosses are all tails. Independence of the coin tosses implies

$$
\mathbb{P}(X \geq x) = \mathbb{P}(\text{first } x-1 \text{ tosses are } \texttt{T}) = (1-p)^{x-1}.
$$

Invoking the result from part (c) and the geometric series formula, we obtain

$$
\mathbb{E}[X] = \sum_{x=1}^{\infty} \mathbb{P}(X \geq x) = \sum_{x=1}^{\infty} (1-p)^{x-1} = \sum_{i=0}^{\infty} (1-p)^{i} = \frac{1}{1-(1-p)} = \frac{1}{p}.
$$